



SWAT4HCLS26 report: OpenMRS as a reference implementation platform for Electronic Health Records and Clinical Decision Support

BioHackathon series:

[Biohackathon SWAT4HCLS 2026](#)

AmsterdamUMC

[OpenMRS](#)

Submitted: 09 Apr 2026

License:

Authors retain copyright and release the work under a Creative Commons Attribution 4.0 International License ([CC-BY](#)).

Published by [BioHackrXiv.org](#)

Andra Waagmeester ¹, **Stephanie “Ace” Medlock** ^{1,2}, **Eric Herman**^{2,3}, **Claude Nanjo**⁴, **Chang Sun** ⁵, and **Ronald Cornet** ¹

1 Department of Medical Informatics, AmsterdamUMC, Location AMC Amsterdam, the Netherlands

2 Stichting OpenElectronicsLab, The Netherlands **3** Commons Caretakers B.V., Amsterdam, The Netherlands

4 University of Utah **1** DACS - Data Department of Advanced Computing Sciences, Faculty of Science and Engineering, Maastricht, the Netherlands

Introduction

The broad goal for the OpenMRS table is: What steps are needed to use OpenMRS as a resource for reference implementations?

This can be divided into 4 levels:

1. Use OpenMRS as a data source, using its available APIs This is the easiest one. I will set up an OpenMRS instance that can be used, and provide the URL, some logins, and some instructions. The goal for anyone working on this will simply be to work on whatever demo they want to work on, using the APIs and the fake data provided by this system.
2. Use OpenMRS running locally, so that you can directly access its database The instructions provided by the OpenMRS community for the newest version (O3) are not bad, and I have some detailed notes about how to do this on Linux (see below). I've never done it on Windows. The goal for anyone working on this will be to come up with instructions for installing the system on Windows, and querying the database (e.g. from Python). This is what is needed to have a standalone application running alongside OpenMRS that uses its database.
3. Add a module to OpenMRS This is the first step toward making modifications to OpenMRS itself. The OpenMRS community provides some “instructions for developers”. The goal for anyone working on this will be to follow those instructions (and improve them where needed) with the goal of adding a simple module (I would suggest “add a clickable URL to the patient summary screen” as a good objective for MyFirstModule). This is what is needed to embed new functionality in OpenMRS.
4. Build OpenMRS from source This is what you need if you want to change something in OpenMRS, rather than just adding something to it. The goal will be to come up with a set of instructions for how it can be built from source... preferably in an executable format.

A secondary objective is defining a roadmap for using OpenMRS as a teaching resource in medical informatics education. The electronic health record (EHR) is *the* central piece of software in front line health care; this implies that understanding how an EHR works is an important part of a medical informatics curriculum. Open source EHRs allow students to look “under the hood” in a way that isn't possible with closed-source products.

Objective

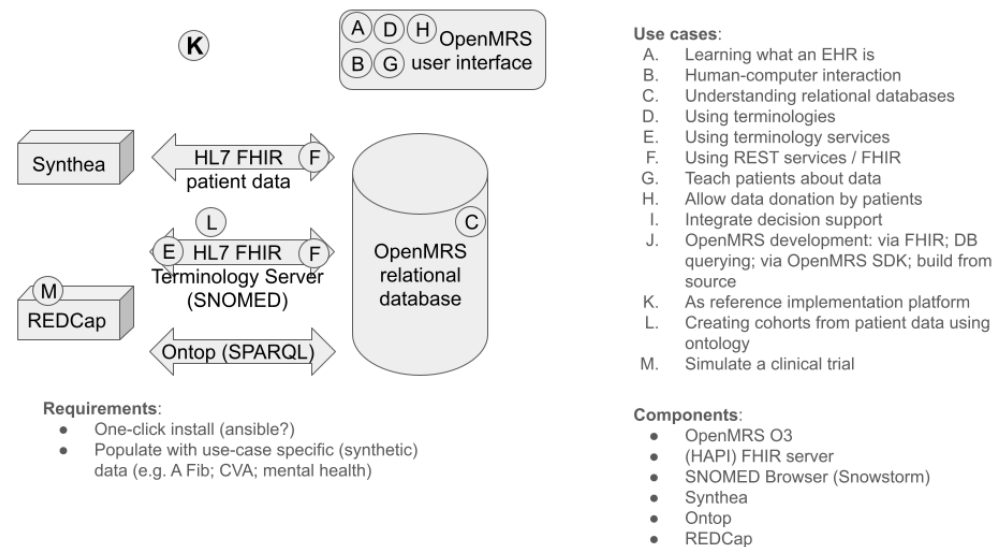


Figure 1: Long term vision

Sources:

- Level 3 (OpenMRS development): https://gitlab.com/openmrs_education/openmrs-sdk-sandbox
- Level 4 (OpenMRS from source): <https://gitlab.com/ericherman/openmrs-sandbox>

Install reference implementation

```
gh repo clone openmrs/openmrs-distro-referenceapplication
cd openmrs-distro-referenceapplication
docker compose up (or docker compose up -d)
```

Waited — saw empty screen / 504s for ~30 min (first-time data load)

```
docker compose restart backend (when stuck)
```

Waited ~17 min for backend to finish loading Accessed <http://localhost/openmrs/spa>, logged in with admin / Admin123

Initial screen after succesful reference implementation

Contributor Role Taxonomy

A last feature since is minimal support for the Contributor Role Taxonomy (CRedit). You can specify the role of authors in writing the report with the `role:` field. However, the authors are responsible for selection the right terms from [CRedit](#). An example looks like this:

authors:

```
- name: First Author
  affiliation: 1
  orcid: 0000-0000-0000-0000
  role: Conceptualization, Writing - review & editing
```



A full examples

A full example then has this structure:

```
authors:
  - name: First Author
    affiliation: 1
    role: Writing - original draft
  - name: Last Author
    orcid: 0000-0000-0000-0000
    affiliation: 2
    role: Conceptualization, Writing - review & editing
affiliations:
  - name: First Affiliation
    index: 1
  - name: ELIXIR Europe
    ror: 044rwnt51
    index: 2
```

Formatting

This document use Markdown and you can look at [this tutorial](#).

Subsection level 2

Please keep sections to a maximum of only two levels.

Tables

Tables can be added in the following way, though alternatives are possible:

Table: Note that table caption is automatically numbered and should be given before the table itself.

```
| Header 1 | Header 2 |
| ----- | ----- |
| item 1 | item 2 |
| item 3 | item 4 |
```

This gives:

Table 1: Note that table caption is automatically numbered and should be given before the table itself.

Header 1	Header 2
item 1	item 2
item 3	item 4

Figures

A figure is added with:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png)
```

This gives:



Figure 2: Caption for BioHackrXiv logo figure

Figures can be scaled by adding the width or height to the Markdown like this:

```
![Caption for BioHackrXiv logo figure](./biohackrxiv.png){ width=50px }
```

Other main section on your manuscript level 1

Lists can be added with:

1. Item 1
2. Item 2

Citation Typing Ontology annotation

You can use [CiTO](#) annotations, as explained in [this BioHackathon Europe 2021 write up](#) and [this CiTO Pilot](#). Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

The syntax in Markdown is as follows: a single intention annotation looks like `[@usesMethod In:Krewinkel2017]`; two or more intentions are separated with colons, like `[@extends:discusses:Nielsen2017Scholia]`. When you cite two different articles, you use this syntax: `[@citesAsDataSource:Ammar2022ETL; @citesAsDataSource:Arend2022BioHackEU22]`.

Possible CiTO typing annotation include:

- `citesAsDataSource`: when you point the reader to a source of data which may explain a claim
- `usesDataFrom`: when you reuse somehow (and elaborate on) the data in the cited entity
- `usesMethodIn`
- `citesAsAuthority`
- `citesAsEvidence`
- `citesAsPotentialSolution`
- `citesAsRecommendedReading`
- `citesAsRelated`
- `citesAsSourceDocument`
- `citesForInformation`
- `confirms`
- `documents`
- `providesDataFor`
- `obtainsSupportFrom`
- `discusses`
- `extends`

- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

...

Acknowledgements

...

References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, 32(14), 2236–2238. <https://doi.org/10.1093/bioinformatics/btw214> [cito:citesAsAuthority]

Author contributions

Andra Waagmeester: Writing – original draft Stephanie “Ace” Medlock: Writing – original draft Eric Herman: Conceptualization Claude Nanjo: Writing – original draft Chang Sun: Conceptualization Ronald Cornet: Writing – original draft